

Objectives

Q. Subtypes of phonology (3)

Q. When the focal folds are tight together the resultant sounds are called **Voiced**.

Q. the sponge like tissues contained within air range is called the **diaphragm**.

Q. Another name of phonemic transcription is **(Broad Transcription)**

Q. the length of the VOT is dependent on the practical measurement is based on the practical measure of **aspiration**

Q. In auditory phonetics we study sound with

Q. In English there are **(24)** consonants

Q. Voiceless stops at syllable initial are **ASPIRATED**

Q. Consonants at the end of syllable are **LONGER**

Q. Voiced stops at syllable initial are **VOICELESS**.

Q. Right Transcription of furiously... Choose from options **'fjɜ:.i-**

Q. Voiceless stops are **ASPIRATED** in onset position

Q. Unaspirated voiceless stop has Voice onset time near **ZERO**

Q. all sound produced..... Help of air compressed by **LUNGS**

Q. Frequency, pitch, and loudness are **PHYSICAL** features.

Q. another name for phonemic transcription is _____.

a) Narrow transcription **b) broad transcription** C) orthographic transcription d) technical transcription

Q. Which of the following branches of phonetic focuses on methodologies and aspect of experimental work on----- speech sounds? **Experimental phonetics**

Q. in English there are ---- consonant types

2 3 **5** 7

Q. the short symbols which are used for indicating suprasegmental----- diacritics **[ɨ]**

Q. noticeable event accruing at regular intervals

a. Tone b. pitch **c. rhythm** d. intonation

Q. In order to describe intonation, we need to analyze the role of a '**stressed syllable**'

Q. /ʒ/ and /a/ are examples of vowel

a.lax b.Tense c.lenis d.forties

Q. How many sub types does phonology has

2 3 4

Q. Phonology is the study of..... Structure of language

Sound Grammar

Q. lenis refers to a sound made with a relatively..... Muscular effort.

Less quick Strong Weak

Q. the velaric airstream mechanism involves an air flame produced by a movement of the back of tongue against the

Hard palat Soft palat

Q. Higher overtone F2 tells about the position of TONGUE.

Q. In IPA small symbols superimposed on segments are called DIACRITICS

Q. Lax sounds are produced with LESS muscular effort.

Q. Optimality theory has stemmed from GENERATIVE phonology.

Q. Human speech production is studied under the branch of phonetics.

Q. While studying vowels on spectrogram overtone F2 tells us about the position of ____

Q. phonemic transcription is enclosed in slashes or oblique lines

Q. fricatives are also known as spirants

Q. A phoneme is the smallest meaningful unit of sound

Q. In typical tone languages linguistic function of phoneme is to change meaning of word.

Q. /b/ /d/ fall under the category of ____

voiced voiceless

Q. Which word used in long vowels barn

Q. When a sound is not primarily articulated and some other sound is added to it known as. Articulation

Q. Intonation refers (very) simply to the variations in the pitch of a speaker's voice (f0).

Q. In order to describe intonation, we need to analyze the role of a 'stressed syllable'.

Q. This term is used to describe some varieties of English (e.g., American) pronunciation in which the /r/ phoneme is found in all its phonological contexts.

Q. The term 'stop' is used in the phonetic classification of consonant sounds on the basis of their manner of articulation.

Q. The North wind and the sun is a folktale taken from (Arne Thompson)

Q. /ɛ:/ and /a:/ are the examples of ... (Tense)

Q. The glottis is defined as the space (vocal folds)

Q. In English /p t k/ at the beginning of a syllable are aspirated

02 Marks Questions

Q. Define the term 'rhotic accent'

A rhotic speaker pronounces the letter "R" in hard and water. In other words, rhotic speakers pronounce "R" in all positions.

Q. Two types of sounds involving aspiration process - 2 marks

Phonetically, aspiration is the result of the vocal cords being widely parted at the time of the articulatory release. In some languages (such as English) aspiration is allophonic while in others (such as Urdu) it is phonemic.

Q. Tone language's

In the case of tone languages, it is usual to identify tones as being a property of individual syllables, whereas an intonational tone may be spread over many syllables. Similarly, in the analysis of English intonation, tone refers to one of the pitch possibilities for the tonic (or nuclear) syllable, a set usually including fall, rise, fall-rise and rise-fall, though others are also suggested by various experts.

Q. feel, father, put describe vowels with tongue position

A low vowel (such as a in "father" or "had") is produced with the tongue relatively flat. You may also be able to feel the [i] as somewhat further forward

Q. pronouncing /u/ tongue position is:

Vowel /u/ is often pronounced with spread lips in this variety of English. It is like the back of the tongue is raised for an /u/ vowel sound creating a secondary (articulation) constriction.

Q. H and L stand for in break indicates 2 marks

One of the most recently developed methods is called the 'tone and break indices' (ToBI). This method is used for the description of intonation (H/L) representing pitch changes and showing pitch accent – phrase accent and boundary by showing tone and break indices. This is based on describing High (H) and Low (L) pitches in a sentence representing pitch changes and showing pitch accent – phrase accent and boundary (of the phrase).

Q. Define relation of F1 and F2 in vowel sound?

Acoustically, vowels are mainly distinguished by the first two formant frequencies F1 and F2; F1 is inversely related to the vowel height (which means that smaller F1 amplitude = higher vowels), and F2 is related to the front or back of the vowels (smaller F2 amplitude = more back vowels).

Q. What does Grey areas indicate in chart

Areas of darker gray represent impossible articulations.

Q. Define sibilant fricative

The fricatives [s, z, ʒ, ʃ] are called sibilant sounds. **Sibilant fricatives** (such as s, ʃ) are strong and clearly audible and **strident fricatives** are weak and less audible (such as θ, f).

03 Marks Questions

Q. No. Of syllables of exist.,pronunciation and enough 3 marks

1. Ex-ist
2. Pro-nun-ci-a-tion
3. En-ough

Q. EXPLAIN the features (Voicing, MoA and PoA)

- **Place of articulation** = alveolar. The narrowing of the vocal tract involves the tongue tip & the alveolar ridge
- **Manner of articulation** = oral stop. (The narrowing is complete -- the tongue is completely blocking off airflow through the mouth. ...)
- **Voicing** = voiced. articulation, phonation, etc.

Q. Stress Rules: (V.V.V. IMP)

2 types of stress are important. Firstly, stress on a syllable within a word (the lexical stress) which changes the grammatical category of a word (compare insult with insult) and also change meaning among other things. On the other hand, stress on a word or certain words in a phrase or sentence. This type of stress (on word(s) within sentences) is called sentence level or prosodic stress. This is, in fact, a change or modification to word level stress in a sentence which is basically a change of 'beat' on certain words in a sentence. Remember that, we create 'rhythm' in spoken language on the basis of stress.

Q. Homograph

It is a group (usually a pair) of words that are spelled the same way, & may or may not be pronounced the same way, although the difference in pronunciation is often just a shift in the accented syllable. There is a whole class of homographs that end in -ate, usually with one being a verb and the other being a noun or an adjective related to it. E.g : "Advocate" can be pronounced with a long "a" sound and mean "to speak or write in support of" "Advocate" can also be pronounced with a short "a" sound and refer to a person who supports or pleads the cause of another.

Q. Supra segmental

‘Supra’ means above (beyond) and ‘segments’ means sounds (such as vowels and consonants) so the term ‘suprasegmental’ means ‘above sounds’. The term suprasegmental was initially invented to refer to aspects of sound such as intonation that did not seem to be the properties of individual segments (i.e. the vowels and consonants of which speech is composed).. There has never been full agreement about how many suprasegmental features are to be found in speech, but pitch, loudness, tempo, juncture, syllable, rhythm and stress are the most commonly mentioned ones.

Q. Phonation types:

- **Voiceless** – when the folds are open apart and the air passing through the glottis freely (/t/ or /p/).
- **Voiced** – when the folds are tight together and there is vibration during the air passage through the glottis (e.g., /b/ or /d/).
- **Creaky voice** – when there is a slight opening in the front and the arytenoid cartilages are tight together, so that the vocal folds can vibrate only at the anterior end (the small opening at the top).
- **Breathy or murmuring sound** – when the vocal folds are apart but still they are vibrating - a breathy voice is like a whisper except voice.

Q. Write a brief note on Fortis and Lenis Consonants.

In linguistics, fortis and lenis, also called tense and lax, refer to consonants pronounced with greater and lesser energy. English has fortis consonants, such as the p in pat, with a corresponding lenis consonant, such as the b in bat. Fortis and lenis consonants may be distinguished by tenseness or other characteristics, such as voicing, aspiration, glottalization, velarization, length, and length of nearby vowels. Fortis and lenis were coined for languages where the contrast between sounds such as p and b does not involve voicing. Fortis and Lenis Consonants These are the terms used in the phonetic classification of consonantal sounds on the basis of their manners of articulation. Fortis refers to a sound made with a relatively strong degree of muscular effort and breath force compared with the other sound (known as lenis). The distinction between tense and lax is used for vowels on the similar lines.

Q. Velaric air stream mechanism

In addition to pulmonic and glottalic airstream mechanisms, there is a third possibility involving velum. Under this mechanism, speech sounds are made by sucking the air (see airstream). This sucking mechanism is used first by babies for feeding and by adult humans in later stages of life for such things as sucking liquid through a straw or drawing smoke from a cigarette (using the back of the tongue against the velum). The basic mechanism for this is the air-tight closure between the back of the tongue and the soft palate, just as if the tongue is then retracted, and the pressure in the oral cavity is lowered and suction takes place. Consonants produced with this mechanism are called clicks. These sounds have a distinctive role in some languages such as Zulu. In English, they may be heard in the ‘tut tut’ (or tsk tsk) sounds, and in a few other contexts.

Q. What are click sounds?

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Q. What is Ingressive stream?

Pulmonic airstream mechanism is the most commonly used mechanism for speech production by human beings. Almost all the sounds we produce in speaking are created with the help of air compressed by the lungs. The

adjective used for this lung-created airstream is ‘pulmonic’: the pulmonic airstream may be ingressive (as in breathing in) but for speaking practically, it is always egressive (speech sounds are produced while pushing the air out). In order to understand this mechanism, we’ll have to analyze the human respiratory system. Under this system, the respiratory muscles set the air in motion. Lungs – the sponge like tissues – contained within air cage called the diaphragm – contract and enlarge lung cavity thus creating the egressive – ingressive actions. This mechanism set an air flow for speech production and human beings produce speech sounds while pushing the air out.

Q. Affricates

An affricate sound is a type of consonant which is made of a plosive followed by a fricative with the same place of articulation (so, it is a mixture of two steps or gestures). For example, /tʃ/ (the voiceless affricate) has /t/ and /ʃ/ as a sound at the beginning and end of the English words church /tʃɜːtʃ/.

Q. phoneme

A phoneme is the smallest meaningful unit of sound (therefore, a smallest unit in phonology) in a language and this meaningful unit of sound is one that will change one word into another word. For example, the difference in both ‘white’ and ‘right’ (ignore spellings here, focus on sounds) is the difference of sounds (w – r) which are phonemes and they have the ability to change meaning

Q. lexical stress

Lexical stress is basically related to the primary stress applied at syllable level (when only one syllable is stressed) that has the ability to change the meaning and the grammatical category of a word as in the case of ‘IMport’ (noun) and ‘imPORT’ (verb).

Q. salient features of phonetic transcription any 3

In a phonetic transcription the transcriber may use the full range of phonetic symbols if these are required; a narrow phonetic transcription is one which carries a lot of fine detail about the precise phonetic quality of sounds, while a broad phonetic transcription gives a more limited amount of phonetic information. Transcription is an important part of the present course based on the objective to understand the sound-symbol correspondence. Transcription is a useful technique for highlighting the phonemes of a language during documentation and description.

Q. Rhythm

Sentence rhythm is another feature of a connected speech. Actually, speech is perceived as a sequence of events in time, and the word rhythm is used to refer to the way events are distributed in time. Obvious example of vocal rhythms is chanting as part of games (for example, children calling words while skipping or cricket crowds calling their favorite team’s name). In conversational speech, the sentence rhythm is a bit complicated, but it is clear that the timing of speech is not random. An extreme view (though a quite common one) is that English speech has a rhythm that allows us to divide it up into more or less equal intervals of time called feet, each of which begins with a stressed syllable: this is called the stress-timed rhythm hypothesis (and languages are divided on the basis of this phenomenon into stress-timed and syllable-timed languages).

Q. Acoustic phonetics.

Acoustic phonetics is related to the study of physical attributes of sounds produced by the vocal tract. It is the branch of phonetics which studies the physical properties of speech sound as transmitted between mouth and ear according to the principles of acoustics (the branch of physics devoted to the study of sound).

Q. define diphthongs

Diphthong is a single vowel consisting of the features of two vowels. Its most important feature is the glide from one vowel quality to another one (so basically it is a glide). The BBC accent of English contains a large number (eight in total) of diphthongs including three ending at /ɪ/ (eɪ, aɪ, ɔɪ – as in words bay, buy and boy), two ending at /ʊ/ (əʊ, aʊ – as in words no and now) and three ending at /ə/ (ɪə, eə, ʊə – as in words peer, pair and poor).

Q. name three branches of phonetics

It has three major branches:

1. articulatory phonetics,
2. acoustics phonetics and
3. auditory phonetics.

Q. phonetic aspiration

Aspiration is a puff of noise made when a consonantal constriction is released and air is allowed to escape relatively freely (e.g., in English /p t k/ at the beginning of a syllable are aspirated). Phonetically, aspiration is the result of the vocal cords being widely parted at the time of the articulatory release. In some languages (such as English) aspiration is allophonic while in others (such as Urdu) it is phonemic. Pronunciation teachers used to make learners of English practice aspirated plosives by seeing if they could blow out a candle flame with the rush of air after p t k – this can, of course, lead to a rather exaggerated pronunciation (and superficial burns). A rather different articulation is used for the so-called voiced aspirated plosives found in many Indian languages (often spelt as ‘bh’, ‘dh’, ‘gh’) where after the release of the constriction the vocal folds vibrate to produce voicing, but are not firmly pressed together; the result is that a large amount of air escapes at the same time, producing a “breathy” quality. It is not necessarily only stops that are aspirated as both unaspirated and aspirated affricates also exist in Urdu.

Q. importance of transcription

Transcription is an important tool in phonetics and phonology. Based on a specific set of symbols, transcription is the writing down of a spoken utterance. In its original meaning, the implied word is converted from one representation (e.g. written text) into another (e.g. phonetic symbols). Transcription exercises are a long-established exercise for teaching phonetics. There are many different types of transcription: the most fundamental division that can be made is between phonemic (broad) and phonetic (narrow) transcription. In the case of the former, the only symbols that may be used are those which represent one of the phonemes of the language, and extra symbols are excluded. In a phonetic transcription the transcriber may use the full range of phonetic symbols if these are required; a narrow phonetic transcription is one which carries a lot of fine detail about the precise phonetic quality of sounds, while a broad phonetic transcription gives a more limited amount of phonetic information. Transcription is an important part of the present course based on the objective to understand the sound symbol correspondence. Transcription is a useful technique for highlighting the phonemes of a language during documentation and description.

Q. discuss phonetics and phonology in linguistics

Phonetics	Phonology
General	Specific
Study of human speech/languages	Study of a particular language
How sounds are articulated	How sounds are contrastive
Speech production, transmission and Reception	Syllables, phrases, rhythm, tone, intonation

Q. what is negative VOT or Difference b/w negative and positive VOT OR What is Positive VOT or three possible types of VOT based on the nature of stop sounds

1. Firstly, simple **unaspirated voiceless stops** have a voice onset time at or near zero. This means that the voicing of a following vowel begins at or near to when the stop is released.
2. The second possibility is when aspirated stops are followed by a vowel: voice onset time is greater than zero called a **positive VOT**. The length of the VOT in such cases is based on the practical measure of aspiration – the longer the VOT, the stronger the aspiration lasts twice as long as that of English - 150ms).
3. The third possibility is when voiced stops have a VOT noticeably less than zero called "**negative VOT**". This would simply mean that the vocal cords start vibrating before the stop is released.

Q. Difference between American n BBC accents

Accent difference must be realized while transcribing for a specific accent (e.g., British vs. US accent) as one language can have many possible transcriptions. So, in order to realize the differences based on a typical accent, we need to know four types of variations among various accents of a language:

- a. Difference in phonological inventories. Compare /strʌt/ vs. /strʊt/.
- b. Difference in phonetic features such as pronouncing /t/ as [ʔ].
- c. Phonological distribution e.g., rhotic vs. non-rhotic accents.
- d. Lexical distribution e.g., /θ/ and /ð/ differences (as England and Wales have /ð/ and Scottish accents have /θ/).

Q. Narrow transcription

In narrow transcription, the sounds are symbolized on the basis of their articulatory/auditory identity, regardless of their function in a language (sometimes called an impressionistic transcription). In a phonetic transcription, the aim

is not to judge the functional significance of sounds, in the context of some languages, but to identify the sounds as such (phonetic variation).

Q. Airstream mechanism

‘Airstream’ is a term used in phonetics for a physiological process which provides a source of energy capable of being used in speech sound production. There are various forms and mechanisms for initiating the air move. The most common is when the air is moved inwards or outwards by initiating air movement involving ‘lungs’ (the pulmonic airstream), which is used for producing the majority of human speech sounds. The ‘glottalic’ airstream mechanism, as its name suggests, uses the movement of the glottis - the aperture between the vocal folds as the source of energy. The third one is the ‘velaric’ airstream mechanism which involves an airflow produced by a movement of the back of the tongue against the velum.

Q. Velarization

In co-articulation, velarisation is a process whereby a constriction in the vocal tract is added to the primary constriction which gives a consonant its place of articulation. More specifically, velarisation is an example of secondary articulation. In the case of English “dark /l/”, the /l/ phoneme is produced with its usual primary constriction in the alveolar region (try speaking this sound). It is like the back of the tongue is raised for an /u/ vowel sound creating a secondary (articulation) constriction. There are more examples, life vs. file (/laɪf/ /faɪl/) clap vs. talc (/klæp/ /tælk/). It is a very common feature of Arabic and is quite important and interesting for acoustic analysis.

Q. VOT

Voice Onset Time (VOT) is a term used in phonetics referring to the point in time at which vocal fold vibration starts in relation to the release of a closure (during the production of plosive sounds). In order to understand VOT, the three types of plosive sounds are to be explained – voiced, voiceless and a voiceless aspirated sound. For example, during the production of a fully voiced plosive (e.g., /b/ or /g/), the vocal folds vibrate throughout; in a voiceless unaspirated plosive (such as /p/ or /t/), there is a delay (or lag) before voicing starts; and, in a voiceless aspirated plosive (e.g., /p^h/ or /t^h/), the delay is much longer, depending on the amount of aspiration. The amount of this delay is called Voice Onset Time (VOT) which in relation to the types of plosive varies from language to language.

Q. Types of stops

1. Voiceless stops (e.g., p, t, k) are aspirated when they are syllable initial (pip, test, kick).
2. Voiced stops (b, d, g) and affricate (dʒ) are voiceless when they are syllable initial (except when immediately preceded by a voiced sound – compare a day with this day).
3. Voiceless stops (p, t, k) are unaspirated after /s/ in words such as spew, stew and skew

Q. Pitch and variables

The three variables of pitch

1. range,
2. height and
3. Direction.

Q. Nasalization.

Nasalization is an articulatory process whereby a sound is made ‘nasal’ (when the air is passing through the nasal cavity) due its adjacent nasal sound (it is an articulatory influence of an adjacent nasal consonant, as in words like

mat or hand). A vowel can also be nasalised in words like man (when /a/ may be articulated with the soft palate lowered throughout), because of the nasal consonants' influence (this is called anticipatory coarticulation).

Q. function of intonation

In many languages, including English, intonation serves a grammatical function, distinguishing one type of phrase or sentence from another. Thus, "Your name is John," beginning with a medium pitch and ending with a lower one (falling intonation), is a simple assertion; "Your name is John?", with a rising intonation (high final pitch), indicates a question.

Q. define stop consonant

'Stop' is often used as if synonymous with plosive. However, some phoneticians use it to refer to the class of sounds in which there is complete closure specifically in the oral cavity. In this case, sounds such as m, n are also stops. More precisely, they are nasal stops. In English, there are nine stops (six oral and three nasal):

	bilabial	alveolar	velar
- Voiced	p	t	k
+Voiced	b	d	g
(nasal) +Voiced	m	n	ŋ

Apart from the above stops, in some varieties of English, the glottal stop /ʔ/ is found as in beaten

[ˈbiʔn]. English voiceless stops (p, t, k) are also aspirated in the beginning of the words such as [phaI, thaI, khaI].

Q. characteristics of phoneme

A phoneme is the smallest meaningful unit of sound (therefore, a smallest unit in phonology) in a language and this meaningful unit of sound is one that will change one word into another word. For example, the difference in both 'white' and 'right' (ignore spellings here, focus on sounds) is the difference of sounds (w – r) which are phonemes and they have the ability to change meaning. Similarly, take another example of 'cat' vs. 'bat' (k – b). Linguists have also defined phoneme as a group or class of sound events having common patterns of articulation.

Q. Which sound is produces when vocal folds are apart?

The position of larynx (also known as sound box) and the vocal folds inside larynx are very important in the description of speech sounds. 'Phonation' is a technical term used for describing the forms of vibration of the vocal folds (or vocal cords) and the process is more commonly known as voicing. The glottis (which is defined as the space between the vocal folds) can assume a number of shapes (such as voiced, voiceless, murmuring and creaky positions). The most common positions of vocal folds are used to describe the two possible features of consonant sounds by considering sounds to be either voiceless with the vocal folds apart (such as /p/ and /t/) or voiced with the folds nearly together so that they will vibrate when air passes between them (such as /b/ and /g/). These glottal states are important in the description of speech sounds in particular languages and in the description of pathological voices. The process of phonation is also known as 'voicing' and laryngeal activity.

Q. Essential requirements of human speech production, any 3 - 3 marks

The most common one is the moving of air by compression of the lungs so that the air is expelled through the vocal tract (this is called a pulmonic airstream - usually an egressive pulmonic one, but occasionally speech is produced while breathing in).

The second one is the glottalic (produced by the larynx with closed vocal folds - it is moved up and down like the plunger of a bicycle pump) and

The last one is called velaric (where the back of the tongue is pressed against the soft palate or velum - making an air-tight seal, and then drawn backwards or forwards to produce an airstream).

Q. English consonants rules, fill in the blanks: - 3 marks {Learn it cz I don't know which words were missing}

The BBC accent of English contains a large number (eight in total) of diphthongs including three ending at /ɪ/ (eɪ, aɪ, oɪ – as in words bay, buy and boy), two ending at /ʊ/ (əʊ, aʊ – as in words no and now) and three ending at /ə/ (ɪə, eə, ʊə - as in words peer, pair and poor).

Q. Define the term fricative.

A fricative consonant is made by forcing air through a narrow gap so that a hissing noise is generated. This may be accompanied by voicing (in which case the sound is a voiced fricative, such as [z] or it may be voiceless as [s]). The quality and intensity of fricative sounds varies greatly, but all are acoustically composed of energy at relatively high frequency. There are several fricative sounds in English, both voiced and voiceless, as in fin [f], van [v], thin [θ], this [ð], sin [s], zoo [z], ship [ʃ], measure [ʒ] and hoop [h]. A distinction is sometimes made between sibilant or strident fricatives. Sibilant fricatives (such as s, ʃ) are strong and clearly audible and strident fricatives are weak and less audible (such as θ, f). BBC pronunciation has nine fricative phonemes: f, θ, s, ʃ, h (voiceless) and v, ð, z, ʒ (voiced).

Q. How would you name following stops sounds according to POA

/p/, /b/, /m/ bilabial

/t/, /d/, /n/ alveolar

/k/, /g/, /ŋ/ Velar

Q. International phonetic association/alphabets

Linguistic phonetics is an approach which is embodied in the principles of the International Phonetic Association (IPA) and in a hierarchical phonetic descriptive framework that provides certain basis for formal phonological theory. Speech, being very complex phenomena and having multiple levels of organization, needs to be explored from different angles. Linguistic phonetics answers a number of questions related to the possible ways of articulatory unified phonetics and phonology and from the perspective of cognitive phonetics focusing on speech production and perception and how they shape languages as a sound systems. The idea is mainly related to the overall ability of human beings to produce sounds (as a community and irrespective of their specific languages) and then the representation of their shared knowledge (as considered by the IPA in its charts) for formal phonetic and phonological theories.

Q. secondary articulation OR SAGS

While discussing vowels, we need to highlight some of the secondary articulation related to vowels. Remember that 'secondary' articulation is an articulatory gesture with a lesser degree of closure occurring at approximately the same time as another (primary) gesture. It is different than co-articulation which is at the same time and of the same

value (taking place as an equal level gesture). Thus it is appropriate to consider four types of secondary articulations in conjunction with vowels because they can usually be described as added vowel-like articulations including; 'palatalization' (adding a high front tongue gesture as in sound /i/), velarization (raising of the back of the tongue), pharyngealization (it is the superimposition of the narrowing of the larynx) and labialization (the addition of lip-rounding).

Q. Aims And objective of studying phonetics and phonology:

Among other things, expertise in phonetics and phonology enable researchers to describe spoken languages which are not yet documented thus proving very important for language documentation and language description. Similarly, it is an important field for typological studies and for cross linguistic comparisons and generalizations in terms of sound systems (languages) and their classifications. understand how human sound is produced; know the physical properties of human sounds.

Q. Define intrinsic and extrinsic allophones.

An allophone of a phoneme which may be explained principally by reference to phonetic features of the environment in which occurs. An example is the dental allophone [n̪] of the /n/ phoneme in English. This only occurs immediately before a dental fricative, for example in a word like tenth.

An allophone of a phoneme which cannot be explained simply by reference to phonetic features of the environment in which it appears. An example is the velarised allophone [ɪ̞] of the /ɪ/ phoneme in many accents of English. This appears in all environments except immediately before a vowel or j. There is nothing common in the environments in which it does occur which could explain the velarisation.

Q. Keeping in mind the allophones. Fill the following blanks.

"A vowel is longest in a/an **open** syllable, next longest in a syllable closed by a/an **voiced** consonant and shortest in a syllable closed by a/an **voiceless** consonant.(eg. Compare , sea, seed, seat or sight, side,site

Q. Two possible forms in connected speech

Words can have two possible forms: weak and strong. In other words, one of the two possible pronunciations for a word in the context of connected speech is 'strong' and the other is 'weak'. The 'strong' form is the result of a word being stressed (e.g., I want bacon and eggs vs. I want bacon and eggs – where the stress is on AND in order to emphasize it). The notion is also used for syntactically conditioned alternatives, such as your book vs. the book is yours. On the other hand, the weak form is that which is the result of a word being unstressed as in the normal pronunciation of OF in cup of tea, and in most other grammatical (closed form of) words. Several (closed class/function) words in English have more than one weak form (e.g. and [ænd] can be [ənd], [ən], [n], etc.

Q. Three stages of history of phonology. 3 marks

Usually in phonetics we are only interested in sounds that are used in meaningful speech, and phoneticians are interested in discovering the range and variety of sounds used this way in all the known languages of the world. This is sometimes known as linguistic phonetics. Thirdly, there has always been a need for agreed conventions for using phonetic symbols that represent speech sounds; the International Phonetic Association has played a very important role in this regard. Finally, the auditory aspect of speech is very important: the ear is capable of making fine discrimination between different sounds, so much so that sometimes it is not possible to define in articulatory terms precisely what the difference is (but we can still hear the difference).

Q. Write a note on glottis:

The space between the vocal folds (glottis - inside the larynx) can assume a number of positions thus modifying the form of speech sounds and changing the features of it. When brought into light this contact of vocal folds with each other, the shape of vibration (of air passage) is forced to produce phonation or voicing. So, based on the possibilities of the nature of vibration, the states of glottis are determined thus the vibration can be made to vary in many ways, resulting in differences in such things as pitch, loudness and voice quality. If a narrow opening is made between the vocal folds, friction noise can result and this is found in whispering and in the glottal fricative /h/. A more widely open glottis is found in most voiceless consonant.

Q. Write any three accents of language from the four.

1. Difference in phonological inventories. Compare /strʌt/ vs. /strʊt/.
2. Difference in phonetic features such as pronouncing /t/ as [ʔ].
3. Phonological distribution e.g., rhotic vs. non-rhotic accents.

Q. Types of phonetics: (IMP)

1. **Articulatory phonetics** deals with studying the making of single sounds by the vocal tract. It is the branch of phonetics which studies the way in which speech sounds are made ('articulated') by the vocal organs.
2. **Acoustic phonetics** is related to the study of physical attributes of sounds produced by the vocal tract. It is the branch of phonetics which studies the physical properties of speech sound as transmitted between mouth and ear according to the principles of acoustics (the branch of physics devoted to the study of sound).
3. **Auditory phonetics** deals with understanding how human ear perceives sound and how the brain recognizes different speech units. This branch of phonetics studies the perceptual response to speech sounds as mediated by ear, auditory nerve and brain.

Q. VOT in Nawajo language

The Navajo aspirated stops have a very large VOT value that is quite exceptional (150 MS). The length of the VOT in such cases is based on the practical measure of aspiration – the longer the VOT, the stronger the aspiration (Navajo, for example, has strongly aspirated stops where, the aspiration (and therefore the VOT) lasts twice as long as that of English - 150ms)

Q. Aspect of connected speech (3):

Overlapping is a common feature of connected speech. In a rapid (connected) speech, overlapping between sounds results in the positions of some parts of the vocal tract being influenced quite a lot by neighboring targets thus creating various forms (allophones) for one phoneme. Keeping in mind this possibility of overlapping, a phoneme is an abstract unit that may be realized in several different ways (forms - allophones). Similarly, the differences between various allophones of a phoneme can be explained in terms of targets and overlapping gestures. The difference between two different forms of /k/ sound (as the [k] in key and the [k] in caw) may be simply due to their overlapping with different vowels in context. Similarly, the alveolar [n] in ten is different than the dental [n̪] in tenth. Both are the result of aiming at the same target, but in tenth, the realization of the phoneme /n/ is influenced by the dental target required for the following sound.

Q. /i/, /u/, /a/, /æ/, describe the position of tongue.3

1. raising of the front of the tongue such as for /i/
2. the back of the tongue is raised for an /u/ vowel sound.
3. /a/ may be articulated with the soft palate lowered throughout
4. low vowel /æ/ as in man and by keeping the soft palate lowered.

Q. accent variation (3 marks)

- Difference in phonological inventories. Compare /strʌt/ vs. /strɒt/.
- Difference in phonetic features such as pronouncing /t/ as [ʔ].
- Phonological distribution e.g., rhotic vs. non-rhotic accents
- Lexical distribution e.g., /θ/ and /ð/ differences (as England and Wales have /ð/ and Scottish accents have /θ/).

05 Marks Questions

Q. American and RP in rhetoricity 5 marks

This term is used to describe some varieties of English (e.g., American) pronunciation in which the /r/ phoneme is found in all its phonological contexts. Remember that in the BBC accent of English, /r/ is only found before vowels (as in 'red' /red/, 'around' /əraʊnd/), but never before consonants or before a pause. In rhotic (e.g., some American) accents, on the other hand, /r/ may occur before consonants (as in 'cart' /kɑːrt/) and before a pause (as in 'car' /kɑːr/). While the BBC accent is non-rhotic, many accents of the British Isles are rhotic (including most of the south and west of England, much of Wales, and all of Scotland and Ireland). Similarly, most speakers of American English speak with a rhotic accent, but there are non-rhotic areas including the Boston area, lower-class New York and the Deep South. From English language teaching point of view, foreign learners encounter a lot of difficulty in learning not to pronounce /r/ in the wrong places.

Q. Rules for ECA

1. Consonants are longer when at the end of a phrase (e.g., bib, did, don and nod).
2. Voiceless stops (e.g., p, t, k) are aspirated when they are syllable initial (pip, test, kick).
3. Voiced obstruents (b, d, g, v, ð, z, ʒ) are voiced only when they occur at the end of an utterance or before a voiceless sound.
4. Voiced stops (b, d, g) and affricate (dʒ) are voiceless when they are syllable initial (except when immediately preceded by a voiced sound – compare a day with this day).
5. Voiceless stops (p, t, k) are unaspirated after /s/ in words such as spew, stew and skew.

Q. short vowels or long vowels or diphthongs or consonants types with examples: OR

Q. Underline the plosive in given words OR Q. how many vowels in english? write four words with short vowels

Vowels : 20	Consonants: 24
Pure vowels: monophthongs: 12: Short vowels 7 • ɪ pit • e pet • æ pat • ʌ putt • ɒ pot • ʊ put • ə another Long vowels 5 • i: bean	Plosive sounds: 6 • p pin • b bin • t tin • d din • k kin • g gum Nasal sounds: 3 • m sum

- ɑ: barn
- ɔ: born
- u: boon
- ɜ: burn

Diphthongs 8

- ɪə peer
- eə pair
- ʊə poor
- eɪ bay
- aɪ buy
- ɔɪ boy
- əʊ no
- aʊ now

- n sun
- ŋ sung

Fricative sounds: 9

- f fine
- v vine
- θ think
- ð this
- s seal
- z zeal
- ʃ sheep
- ʒ measure
- h how

Affricate sounds: 2

- tʃ chain
- dʒ Jane

Approximant sounds: 4

- l light
- r right
- w wet
- j yet

Q. Three major areas of speech production in articulatory phonetics or what is velar or bilabial or any one of them. [It can be a 2 or 3 or 5 marks question]

- **Bilabial:** This sound is made with two lips (for example, /p/ and /b/). The lips come together for these sounds.
- **Labiodental:** This sound is made when the lower lip is raised to touch the upper front teeth (for example, /f/ and /v/).
- **Dental:** This sound is made with the tongue tip or blade and upper front teeth. For example, say the words thigh, thy and you will find the first sound in each of these words to be dental.
- **Alveolar:** This sound is made with the tongue tip or blade and the alveolar ridge. You may pronounce words such as tie, die, nigh, sigh, zeal, lie using the tip of the tongue or the blade of the tongue for the first sound in each of these words (which are alveolar sounds).
- **Retroflex:** This sound is produced when the tongue tip curls against the back of the alveolar ridge. Many speakers of English do not use retroflex sounds at all but it is a common sound in Pakistani languages such as Urdu, Sindhi, Pashto, Balochi and Punjabi.
- **Palato-alveolar:** This sound is produced with the tongue blade and the back of the alveolar ridge (for example, first sound in each of words like shy, she, show)
- **Palatal:** This sound is produced with front of the tongue and the hard palate (such as the first sound in 'yes').
- **Velar:** This sound is produced with back of the tongue and the soft palate (such as /k/ and /g/).

Q. define tap and flap 5mrks

Tap	Trill	Flap
Tap is up and down movement of the top of the tip of tongue. For example, pronouncing the middle sound in word 'pity' with typical American accent [ɾ]. It is very brief and is produced by a sharp upward throw of the tongue blade. In this	In the production of trill the articulator is set in motion by the current of air [r]. It is a typical sound of Scottish English as in words like 'rye' and 'row'.	Flap is front and back movement of tongue tip at the underside of tongue with curling behind. It is found in abundance in Indo-Aryan (IA) languages [ɾ]. Typical flap sounds found in IA languages is a retroflex sound and the examples are [ɽ], [ɖ]

sound, tongue makes a single tap against the alveolar ridge.

and [ŋ].

Q. Features of connected speech. 5 marks

This type of transcription is required when a spoken language is analyzed as a continuous sequence, as in normal utterances and conversations. It is now realized by experts that important changes happen to sound units when they are used in a connected speech e.g., and becoming /n/ in such phrases as boys and girls. One more example is /n/ becoming /m/ in the phrase green bus. The features of connected speech are: assimilation, rhythm, stress, elision, linking, tone and intonation.

Q. Examples of Velaric airstream mechanism (5)

In addition to pulmonic and glottalic airstream mechanisms, there is a third possibility involving velum. Under this mechanism, speech sounds are made by sucking the air (see airstream). This sucking mechanism is used first by babies for feeding and by adult humans in later stages of life for such things as sucking liquid through a straw or drawing smoke from a cigarette (using the back of the tongue against the velum). The basic mechanism for this is the air-tight closure between the back of the tongue and the soft palate, just as if the tongue is then retracted, and the pressure in the oral cavity is lowered and suction takes place. Consonants produced with this mechanism are called clicks. These sounds have a distinctive role in some languages such as Zulu. In English, they may be heard in the 'tut tut' (or tsk tsk) sounds, and in a few other contexts.

Q. Vowel position at Center, end, start:

One of the major problems describing vowels is the difficulty to describe precisely the tongue position (during the production of a vowel) as people cannot determine appropriately for themselves where their tongues are. So it is important for you to remember that the terms we are using (for the description of vowels) are simply labels that describe how vowels sound in relation to one another. They are not absolute descriptions of the position of the body of the tongue. The reason is that it is perfectly possible to make a vowel sound that is halfway between a high-vowel and a mid-vowel and even it is possible to make a vowel at any specified distance between any two other vowels. This is because of the fact that vowels form a continuum (try gliding from one vowel to another -from /æ/ in had to /i/ as in he (try to stay as long as possible on the sounds between them). The result you can see is the difference in vowel quality.

Q. Difference between diphthongs and monophthongs. Or Q. Write a note on pure vowels? 5

Diphthong is a single vowel consisting of the features of two vowels. Its most important feature is the glide from one vowel quality to another one (so basically it is a glide). The BBC accent of English contains a large number (eight in total) of diphthongs including three ending at /ɪ/ (eɪ, aɪ, ɔɪ – as in words bay, buy and boy), two ending at /ʊ/ (əʊ, aʊ – as in words no and now) and three ending at /ə/ (ɪə, eə, ʊə – as in words peer, pair and poor). There had been a point of difference whether a diphthong should be treated as a single phoneme (in its own right) or it is a combination of two phonemes. On the basis of phonetic classification of vowel sounds and manners of articulation, we need to compare diphthongs with monophthongs and triphthongs; - a monophthong is a vowel with no qualitative change in it – a diphthong is a vowel where there is a single (perceptual) noticeable change in quality during a syllable (as in English words beer, time and loud) - a triphthong is a vowel where two such changes can be heard.

Q. How Wavesform are helpful in Consonants identification... 5 Marks

As part of the acoustics of consonants, we need to note down a few distinctive points about their waveforms. Although the places of articulation are not obvious in any waveform yet the differences in some of the principal manners of articulation—stop, nasal, fricative, and approximant—are usually apparent. Furthermore, as already pointed out, we can also see the differences between voiced and voiceless sounds. The difference between a vowel and a consonant in the waveform is very easy to understand. For the vowel sounds, the lips open and the amplitude gets larger. We can also observe the time duration of a sound from its waveform (vowels are longer than consonants). In case of a stop sound, the closure and the burst are also easy to judge. Similarly, we can also note down the voicing bar for a ‘voiced’ sound producing small voicing vibrations instead of a flat line. A fricative with a more nearly random waveform pattern can also be judged from its waveform.

Q. What is lexical stress and give characteristics of stressed syllable.5 marks

Lexical stress is basically related to the primary stress applied at syllable level (when only one syllable is stressed) that has the ability to change the meaning and the grammatical category of a word as in the case of ‘IMport’ (noun) and ‘imPORT’ (verb). ‘Stress timed languages’ is a very general phrase used in phonetics to characterize the pronunciation of languages displaying a particular type of rhythmic pattern that is opposed to that of syllable-timed languages. In stress-timed languages, it is claimed that the stressed syllables recur at regular intervals of time (stress-timing) regardless of the number of intervening unstressed syllables as in English. This characteristic is sometimes also referred to as ‘isochronism’, or isochrony. However, it is clear that this regularity is the case only under certain conditions, and the extent to which the tendency towards regularity in English is similar to that in, say, other Germanic languages remains unclear. In short, the division among the syllables is made on the basis of stress and unstressed patterns. In such languages, stress is realized both at word and sentence levels approximately changing the rhythmic patterns (particularly at sentence level).

Q. Write 2 sentences with closed words where that closed words can be used as Stressed... 5 Marks

The citation speech/conversational speech difference is particularly noticeable for the closed class of words. This class of words such as determiners (a, an, the), conjunctions (and, or), and prepositions (of, in, with)—the grammatical words—are very rarely emphasized in the connected speech, and thus their normal pronunciation in the connected speech is quite different from their citation speech forms. But remember that, as with other words, closed-class words show a strong form, which occurs when the word is emphasized, as in sentences such as:

1. He wanted pie and ice cream, not pie or ice cream.
2. There is also a weak form which occurs when the word is in an unstressed position.

Q. lax and tense vowels...

Lax vowels, remember, are short. Tense vowels are also called long vowels; this name is slightly misleading because, in RP English at least, the tense vowels have variable length; they can be much longer than the lax vowels, but under certain conditions they become clipped, or shortened to roughly lax vowel length.

Tense vowels, then, have more variable length than lax vowels. Tense vowels can be unchecked; that is, they can occur at the end of a word. Lax sounds are produced with less muscular effort and movement, and are relatively short and indistinct vowel sounds (e.g., i, e, ɒ, æ, ʌ, ʊ, ə vowels articulated near the center of the vowel area) compared to tense sounds (e.g., u:, i:, ɜ:, a:, ʊə, iə). In other words, a lax vowel is said to be the one produced with relatively little articulatory energy.

Q. allophones OR Q. What does the statement "It is systematic variant of phoneme" tells.

An allophone is a definable systematic variant of a phoneme. Compare the following sets:

1. 's' sound in words like sill, still and spill or in words like seed, steed and speed
2. 'k' sound in words like, key and car
3. 't' sound in words like true and tea
4. 'n' sound in words like tenth and ten
5. Phone is a sound pattern having some acoustic features.

Q. Main 5 focuses of phonology. 5 marks

It focuses on the organization of sounds by studying speech patterns (e.g., phonological rules within a specific language). The key words for describing phonology are 'distribution' and 'patterning' related to speech. Phonologists may look into questions like – why there is a difference in the plurals of cat and dog; the former ends with an 's' sound, whereas the latter ends with the 'z' sound. Phonetics, on the other hand, is the study of actual process of sound making. Phonetics has been derived from the Greek word 'phone' meaning sound or voice. It covers the domain of speech production and its transmission and reception. The sounds made by us when we talk are studied through different branches of phonetics like acoustic phonetics, auditory phonetics and articulatory phonetics.

Q. Mention the five points of acoustic phonology suggested by the phoneticians.

1. Acoustic phonetics is related to the study of physical attributes of sounds produced by the vocal tract.
2. It is the branch of phonetics which studies the physical properties of speech sound as transmitted between mouth and ear according to the principles of acoustics (the branch of physics devoted to the study of sound).
3. It is primarily dependent on the use of instrumental techniques of investigation (such as Praat software), particularly electronics, and some grounding in physics and mathematics is a prerequisite for advanced study of this subject.
4. Its importance to the phonetician is that acoustic analysis can provide a clear, objective datum for the investigation of speech – the physical 'facts' of speech sounds (such as duration, formants F1, F2 and F3, etc.).
5. Thus, acoustic evidence is often referred to when one wants to support an analysis being made in articulatory or auditory phonetic terms.

Q. Compound words, long words in detail.

In linguistics, a compound is a lexeme (less precisely, a word or sign) that consists of more than one stem. Compounding, composition or nominal composition is the process of word formation that creates compound lexemes. That is, in familiar terms, compounding occurs when two or more words or signs are joined to make one longer word or sign. The meaning of the compound may be similar to or different from the meaning of its components in isolation. The component stems of a compound may be of the same part of speech—as in the case of the English word footpath, composed of the two nouns foot and path—or they may belong to different parts of speech, as in the case of the Eng word blackbird, composed of the adjective black and the noun bird. With very few exceptions, English compound words are stressed on their first component stem. In long words (such as minimization), one syllable receives primary stress while other (1 or more than 1 syllable) has the secondary stress. Secondary stress is just for better pronunciation (not changing the meaning). Another division is made between lexical stress (phonemic in nature) and sentence level stress.

Q. Enlist the ideas of generation phonology as presented by Chomsky and Halle?

Like other areas of grammar, a major change in the theory of phonology came about in the 1960s when many people became convinced that important facts about the sound systems of languages were being missed by phonologists

who concentrated solely on the identification of phonemes and the analysis of relationships between them. Morris Halle and Noam Chomsky showed that there were many sound processes which, while they are observable in the phonology, are actually regulated by grammar and morphology. This area of phonology is mainly related to specific phonological rules within languages. These rules describe how they work and regulate changes such as substitutions, deletions and insertions of sounds in specific contexts. In order to highlight these phonological rules, an elaborate method of writing in an algebra-like style was evolved: this can be seen in the best known generative phonological treatment of English (The Sound Pattern of English by Chomsky and Halle, 1968). This type of phonology became extremely complex; it has now been largely replaced by newer approaches to phonology, many of which, despite rejecting the theory of the Sound Pattern of English, are still classed as generative since they are based on the principle of an abstract, underlying phonological representation of speech which needs rules to convert it into phonetic realizations.